

Question				Marking details	Marks available																				
					AO1	AO2	AO3	Total	Maths	Prac															
6	(a)	(i)		<table><tr><th>Compound</th><th>Flame test colour</th><th>Symbol of ion</th><th>Observation on adding silver nitrate solution</th><th>Symbol of ion</th></tr><tr><td>S</td><td>brick red</td><td>Ca²⁺</td><td>yellow precipitate</td><td>I⁻</td></tr><tr><td>T</td><td>apple green</td><td>Ba²⁺</td><td>white precipitate</td><td>Cl⁻</td></tr></table> award (1) mark for each correct answer do not accept ions with missing (or incorrect) charges but penalise once only e.g. award (2) if all three ions identified but no charges given award (1) if two ions identified but no charges given	Compound	Flame test colour	Symbol of ion	Observation on adding silver nitrate solution	Symbol of ion	S	brick red	Ca ²⁺	yellow precipitate	I ⁻	T	apple green	Ba ²⁺	white precipitate	Cl ⁻	4			4		4
Compound	Flame test colour	Symbol of ion	Observation on adding silver nitrate solution	Symbol of ion																					
S	brick red	Ca ²⁺	yellow precipitate	I ⁻																					
T	apple green	Ba ²⁺	white precipitate	Cl ⁻																					
		(ii)		award (1) for any of following the yellow would hide the white precipitate it would look (pale) yellow (like iodide) / cream (like bromide) there would be a mixture of two precipitates it would not be possible to distinguish between the colours			1	1		1															
	(b)			Ag⁺(aq) + Cl⁻(aq) → AgCl(s) award (1) for ions award (1) for product formula award (1) for state symbols – can only be awarded if ions and product are correct		3		3																	
				Question 6 total	4	3	1	8	0	5															